Thomas Pickell

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RESEARCH INTERESTS

Seeking a PhD position in RF circuit design. My experience includes writing embedded software, high-frequency design and characterization, and antenna design. I plan to leverage this background in FPGA development and THz systems to address the growing challenges in next-generation wireless communications.

EDUCATION

2022-present Undergraduate Student, Rice University
Major In Electrical and Computer Engineering
GPA: 3.62/4.00
Expected Graduation: May 2026

RESEARCH EXPERIENCE

2025-present Research Intern, Improving ADC data throughput and processing with FPGA
Professor Bert Hochwald
Notre Dame Wireless Institute, South Bend, IN

- Architected a high-speed data acquisition system on a BeagleV-Fire FPGA, designing a custom Verilog module to pack ADC data and maximize AXI4 bus efficiency.
- Developed bare-metal C drivers to configure the Memory Protection Unit (MPU) and control DMA operations, enabling the FPGA to write directly to DDR memory.
- Verified system functionality through a memory-to-memory DMA test and diagnosed a hardware-level handshake issue in a more advanced test using a systematic debugging approach.

2025-present Research Assistant, Testing and Imaging with 3.4THz Radar
Professor Henry Everitt, Department of Electrical and Computer Engineering
Rice University, Houston, TX

- Authored a comprehensive 27-page Standard Operating Procedure (SOP) for a complex QCL terahertz radar system, detailing all procedures from cryogenic setup to shutdown.
- Managed the integration and operation of cryogenic and high-frequency electronic subsystems, including configuring vacuum pumps , power supplies , and a RIGOL arbitrary waveform generator to achieve stable laser operation.
- Performed signal verification and troubleshooting using an oscilloscope and spectrum analyzer , developing procedures to optimize the signal via optical alignment

2024 Research Assistant, Metasurface Applications for Terahertz Uplink
Professor Edward Knightly, Rice Networks Group
Rice University, Houston, TX

- Investigated the use of a randomized, programmable metasurface to determine the absolute 2D position of a terahertz (THz) receiver.
- Conducted experiments using a THz pulse transmitter and receiver, varying the receiver's position with linear stages across a 200mm coarse-grain and a 50mm fine-grain area.
- Developed a data pipeline to process raw time-domain signals into a Fourier coefficient codebook, correlating unique signal fingerprints to physical locations.

TECHNICAL PROJECT

2025-present Senior Design Project, Open-Source Phased Array Beamformer
Rice University, Houston, TX

- Design, simulation, and fabrication of an 8-element phased array daughter board for the 2.5-3.5 GHz band, compatible with multiple SDR platforms.
- Focusing on RF circuit layout, simulation and testing/controls software to create a low-cost, open-source beamforming option for academic research.
- Project Repository: https://github.com/yilmaz-kaan/beamers

PRESENTATIONS

2024 "2D Position Estimation using Randomized Programmable Metasurface" at the Rice OURI Summer Symposium
Houston, TX *(August 2)*
2024 "Rice Robotics Club: Jumping Leg" at the Huff Engineering Design Showcase
Houston, TX *(April 11)*
2025 "FireHound: Wide-bandwidth Distributed Spectrum Sensing Platform" at the Notre Dame Summer Undergraduate Research Symposium *(July 23)* Notre Dame, IN ,SpectrumX Center Meeting Poster Session *(October 20),* Madison, WI

TEACHING EXPERIENCE

Fall 2025: Course Assistant, Fundamentals of Electrical Engineering, Rice University

- Clarify core concepts in signal processing, Fourier analysis, and communication theory for ~90 undergraduates during weekly office hours.
- Attend meetings to relay student feedback to the instructor, as well as evaluate exam materials to ensure students would be challenged in the course.

Fall 2024-2025: Teaching/Lab Assistant, Digital Logic Design, Rice University

- Host weekly office hours to help classes of ~70 undergraduate students to understand concepts in CMOS circuit design, boolean algebra, and practical timing analysis
- Assist students in completing FPGA labs, where students are tasked with designing a 16-bit CPU based on a set of specifications given in the class

Fall 2024: Technical Writing Mentor, Engineering Design, Rice University
- Worked with teams of 3-5 undergraduates to improve their technical documentation skills
- Hosted weekly review sections to ensure technical writing was precise and useful
- Mentorship led to surveyed students reporting increased confidence in their technical writing skills

TECHNICAL SKILLS

Programming: Linux, Python, Verilog, C, C++
EDA/RF Tools: FPGA Design, KiCAD, Keysight ADS
Frameworks: PyTorch, Django, Tensorflow,

RELEVANT COURSEWORK

Digital Logic Design
Intro to Computer Vision
2D Materials and Devices (Graduate Level)
Learning from Sensor Data
Algorithms, Complexity, and Approximation
Microwave Engineering
Modern Communication Theory and Practice

LANGUAGES

English (native reading, speaking, written)
Korean (intermediate reading, basic spoken, intermediate written)

REFERENCES

Henry Everitt
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